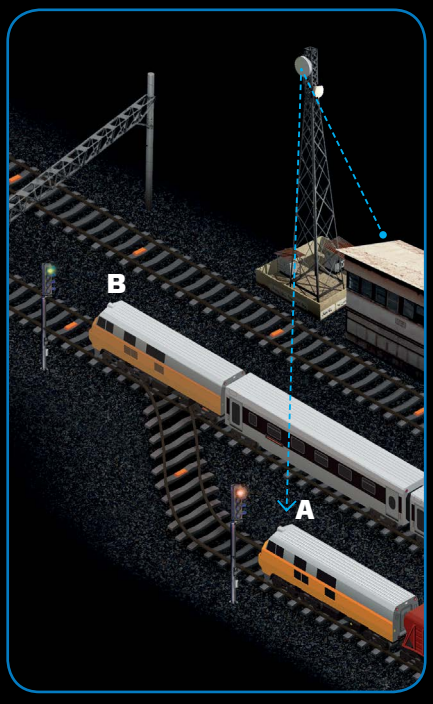
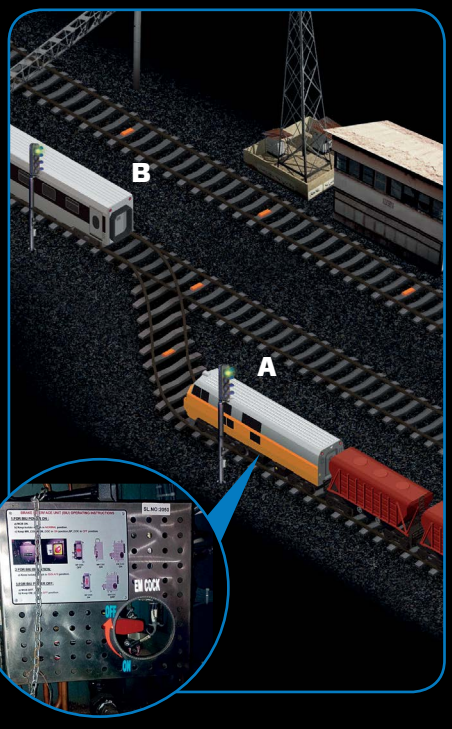


3 **Onboard equipment** on the trains give the loco pilots signalling and movement instructions sent from the ground station



4 Kavach tells A to slow down and **sends an alert** for halting at the red signal so that B can pass along safely



5 If A fails to stop at the red signal, the **automatic braking system** activates and stops the train. Once B is through and at a safe distance, A proceeds

Graphic by **TANMOY CHAKRABORTY**

them in India because there is more rollout now," says Reddy. As of now, the cost of trackside and station equipment of Kavach comes up to approximately Rs 50 lakh per km, and the equipment in the locomotive costs around Rs 70 lakh per loco. In comparison, the ATP system in Europe costs around Rs 2 crore per km.

THE INITIAL BOTTLENECKS

Despite these strengths, there were several hiccups in the implementation of Kavach. On paper, the project stands sanctioned for 35,736 route km of 'High Density' (HDN) and 'High Utilised' (HUN) networks of the railways. But that is phase-II, and it's a long way off. Those associated with the project point to a number of reasons for the delayed implementation, not least of which is the fact that the government could not make up its mind for years initially. As late as 2017, the railway ministry added to the policy confusion by issuing a directive that 10,000 km of routes would get ETCS Level 1—which is inferior to Kavach—and sought rollout plans from zonal railways. "There were also deliberations on whether India should get ETCS Level 2. Anyway, the railways finally decided to go with the indigenous Kavach," says Mahesh Mangal, former senior executive director at RDSO. Reddy offers another reason. In an operational railways, he says, retrofitting takes time. "You can't shut down all traffic at once to instal a system," he adds. Fortunately, funding was never a problem through all of this. "Laying a new railway line costs crores per kilometre, whereas installing Kavach to it adds just a fraction of that to the overall cost," says Mangal. "The real issue was deciding what kind of ATP system the railways wanted to go for."

In the next five years, the Centre plans to instal Kavach on 6,000 km of the busiest routes (over the already in-progress Delhi-Howrah and Delhi-Mumbai ones). In fact, the railway manual states that wherever the sanctioned train speed is over 140 kmph, in-cab signalling is a must. "As the capability of Indian engineers enhances over time, the rollout would be as fast as 5,000 km a year," says Vaishnaw. The railways also plans to export the Kavach technology globally in future. "Not just India, but even other nations can benefit from Kavach. It is cheap and effective, and it has finally broken the stranglehold of international companies, which for years wanted to push foreign ATPs," says Sunil Kumar, former advisor (safety) in the railways ministry.

Loco pilots have welcomed this renewed momentum. They have a word of caution, though. "The train protection systems used in metro systems are similar to Kavach, but safer. There, the train is not allowed to jump a red signal," says K.C. James, secretary-general of the All India Loco Running Staff Association. "In Indian Railways, a train travels far and wide, and running conditions are not ideal, as in trials.... Kavach is good, but, in India, accidents happen due to all kinds of reasons." Kavach, per James, should be "100 per cent safe". Surely not too much of an ask for a network that ferries close to 22 million passengers daily. ■